MATH-101 Calculus and Analytical Geometry – 3-0 CHs

By

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DOT PRODUCT
&
CROSS PRODUCT

Lec # 3

OBJECTIVE

- In this lecture we will define a new kind of multiplication in which two vectors are multiplied to produce a scalar. This multiplication operation has many uses, some of which we will also discuss in this lecture.

- In many applications of vectors in mathematics, physics, and engineering, there is a need to find a vector that is orthogonal to two given vectors. In this lecture we will discuss a new type of vector multiplication that can be used for this purpose.

DOT PRODUCT

INTRODUCTION

Two useful concepts that involves vectors $\vec{A}$ and $\vec{B}$ are the dot product, which is a scalar, and the vector product, which is a vector.
Now first discuss the Scalar products

DOT PRODUCT / SCALAR PRODUCT

The dot product $\vec{A} \cdot \vec{B}$ of two vectors may be defined geometrically or algebraically.

Definition 1

Geometric Definition:

When two non-zero vectors $\vec{A}$ and $\vec{B}$ are placed so their initial points coincide,

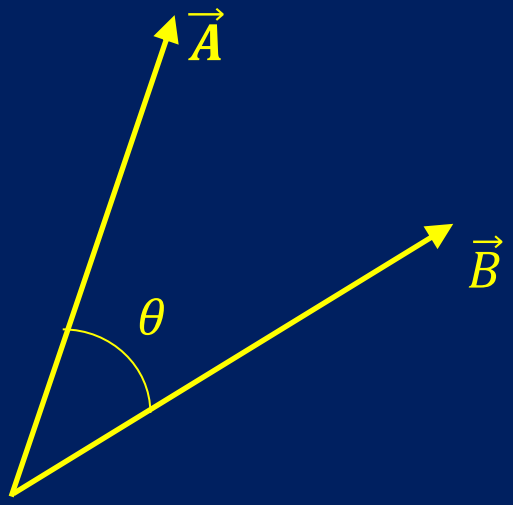
they form an angle θ of measure $0 \leq \theta \leq \pi$. This angle is called the angle b/w $\vec{A}$ and $\vec{B}$.

Definition:

The Scalar product (Dot product) $\vec{A} \cdot \vec{B}$ of vectors $\vec{A}$ and $\vec{B}$ is the number

$$\vec{A} \cdot \vec{B} = |\vec{A}||\vec{B}|cos\theta$$ → Eq. (1)

Where θ is the angle b/w $\vec{A}$ and $\vec{B}$



Definition 2

Algebraic Definition/Dot Product in component form:

To find the scalar product of two given vectors, we multiply their corresponding components $\hat{\imath}, \hat{\jmath}$ *and* $\hat{k}$ and add the results.

If $\vec{A} = a_1\hat{\imath} + a_2\hat{\jmath} + a_3\hat{k}$ and

$\vec{B} = b_1\hat{\imath} + b_2\hat{\jmath} + b_3\hat{k}$ then

$$\vec{A} \cdot \vec{B} = (a_1\hat{\imath} + a_2\hat{\jmath} + a_3\hat{k}) \cdot (b_1\hat{\imath} + b_2\hat{\jmath} + b_3\hat{k})$$

$$\vec{A} \cdot \vec{B} = a_1b_1 + a_2\,b_2 + a_3b_3 \longrightarrow \text{Eq. (2)}$$

Solving Eq. (1) for θ gives a formula for finding angles between vectors

The angle between two non-zero vectors $\vec{A}$ and $\vec{B}$ is

$$\theta = \cos^{-1}\left(\frac{\vec{A} \cdot \vec{B}}{|\vec{A}||\vec{B}|}\right) \longrightarrow \text{Eq. (3)}$$

Example:

Find the angle between the vectors

$\vec{A} = 4\hat{\imath} - 3\hat{\jmath} + \hat{k}$ and $\vec{B} = -\hat{\imath} - 2\hat{\jmath} + 2\hat{k}$

Solution:

$$\vec{A} \cdot \vec{B} = 4(-1) + (-3)(-2) + (1)(2) = -4 + 6 + 2 = 4$$

$$|\vec{A}| = \sqrt{(4)^2 + (-3)^2 + (1)^2} = \sqrt{16 + 9 + 1} = \sqrt{26}$$

$$|\vec{B}| = \sqrt{(-1)^2 + (-2)^2 + (2)^2} = \sqrt{1 + 4 + 4} = \sqrt{9} = 3$$

$$\theta = \cos^{-1}\left(\frac{\vec{A} \cdot \vec{B}}{|\vec{A}||\vec{B}|}\right) = \cos^{-1}\left(\frac{4}{3\sqrt{26}}\right) = 1.31 \text{ radian}$$

$$= 74.84^0$$

LAWS OF DOT PRODUCT/ PROPERTIES OF DOTPRODUCT

1. Commutative Law:

The dot product is commutative

If $\vec{A} = a_1\hat{\imath} + a_2\hat{\jmath} + a_3\hat{k}$ and

$\vec{B} = b_1\hat{\imath} + b_2\hat{\jmath} + b_3\hat{k}$ then

$$\vec{A} \cdot \vec{B} = (a_1\hat{\imath} + a_2\hat{\jmath} + a_3\hat{k}) \cdot (b_1\hat{\imath} + b_2\hat{\jmath} + b_3\hat{k})$$

$$\vec{A} \cdot \vec{B} = a_1b_1 + a_2\,b_2 + a_3b_3 = b_1a_1 + b_2a_2 + b_3\,a_3$$

$$\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A} \qquad \text{Eq. (4)}$$

2. Distributive Law:

The dot product obey the distributive law

$$\vec{A} \cdot (\vec{B} + \vec{C}) = \vec{A} \cdot \vec{B} + \vec{A} \cdot \vec{C} \quad \text{Eq. (5)}$$

Or

$$(\vec{A} + \vec{B}) \cdot \vec{C} = \vec{A} \cdot \vec{C} + \vec{B} \cdot \vec{C} \quad \text{Eq. (6)}$$

If $\vec{A} = a_1\hat{\imath} + a_2\hat{\jmath} + a_3\hat{k}$,

$\vec{B} = b_1\hat{\imath} + b_2\hat{\jmath} + b_3\hat{k}$ and

$\vec{C} = c_1\hat{\imath} + c_2\hat{\jmath} + c_3\hat{k}$ then

$$\vec{B} + \vec{C} = (b_1 + c_1)\hat{\imath} + (b_2 + c_2)\hat{\jmath} + (b_3 + c_3)\hat{k}$$

$$\vec{A} \cdot (\vec{B} + \vec{C}) = (\mathbf{a_1\hat{\imath} + a_2\hat{\jmath} + a_3\hat{k}}) \cdot ((\boldsymbol{b_1 + c_1})\hat{\imath} + (\boldsymbol{b_2 + c_2})\hat{\jmath} + (\boldsymbol{b_3 + c_3})\hat{k})$$

$$= a_1(b_1 + c_1) + a_2(b_2 + c_2) + a_3(b_3 + \boldsymbol{c_3})$$

$$= (a_1 b_1 + a_2 b_2 + a_3 b_3) + (a_1 c_1 + a_2 c_2 + a_3 c_3)$$

$$= \vec{A} \cdot \vec{B} + \vec{A} \cdot \vec{C}$$

REMARK:

Equation (5) and Equation (6) together permit us to multiply sums of vectors by the familiar laws of algebra. For Eq.

$$(\vec{A}+\vec{B})\cdot(\vec{C}+\vec{D})=\vec{A}\cdot\vec{C}+\vec{A}\cdot\vec{D}+\vec{B}\cdot\vec{C}+\vec{B}\cdot\vec{D}$$

3. Let $\hat{A}$ be a unit vector, then $\hat{A}\cdot\hat{A}=1$, Thus

$$\hat{\imath}\cdot\hat{\imath}=1,\ \hat{\jmath}\cdot\hat{\jmath}=1,\ \hat{k}\cdot\hat{k}=1$$

$$\hat{\imath}\cdot\hat{\jmath}=0,\hat{\jmath}\cdot\hat{k}=0,\hat{k}\cdot\hat{\imath}=0$$

4. $\vec{A}\cdot\vec{A}=|\vec{A}|\,|\vec{A}|\cos\theta=|\vec{A}|^2=A^2$, So the square of a vector is equal to the square of its modulus.

PREPENDICULAR (ORTHOGONAL) VECTORS:

Two vectors $\vec{A}$ and $\vec{B}$ are perpendicular $\Leftrightarrow$ (if and only if)

$$\vec{A} \cdot \vec{B} = 0$$

PROOF:

Let $\vec{A}$ be perpendicular to $\vec{B}$, then an angle θ between them is 90^0.

Therefore

$$\vec{A} \cdot \vec{B} = |\vec{A}|\,|\vec{B}|\, cos\ 90^0 = 0$$

$$\vec{A} \cdot \vec{B} = 0$$

Conversely:

Let $\vec{A} \cdot \vec{B} = 0$ then

$$|\vec{A}|\,|\vec{B}|\, cos\theta = 0$$

$$cos\theta = 0$$

$$\Rightarrow \theta = 90^0$$

This shows that $\vec{A}$ is perpendicular to $\vec{B}$

REMARK:

If $\vec{A} \cdot \vec{B} = 0$, vectors $\vec{A}$ and $\vec{B}$ are called orthogonal.

EXAMPLE: Prove that the vectors are orthogonal

a) $\hat{\imath}$, $\hat{\jmath}$

b) $3\hat{\imath} - 7\hat{\jmath} + 2\hat{k}$ and $10\hat{\imath} + 4\hat{\jmath} - \hat{k}$

SOLUTION:

a) $\hat{\imath} \cdot \hat{\jmath} = 0$ (Understood)

b) $(3\hat{\imath} - 7\hat{\jmath} + 2\hat{k}) \cdot (10\hat{\imath} + 4\hat{\jmath} - \hat{k})$

$= 30 - 28 - 2 = 0$

EXAMPLE:

Show that the vectors

$\vec{a} = 3\hat{\imath} - 2\hat{\jmath} + \hat{k}$, $\vec{\mathrm{b}} = \hat{\imath} - 3\hat{\jmath} + 5\hat{k}$, and

$\vec{\mathrm{c}} = 2\hat{\imath} + \hat{\jmath} - 4\hat{k}$ from a right-angle triangle.

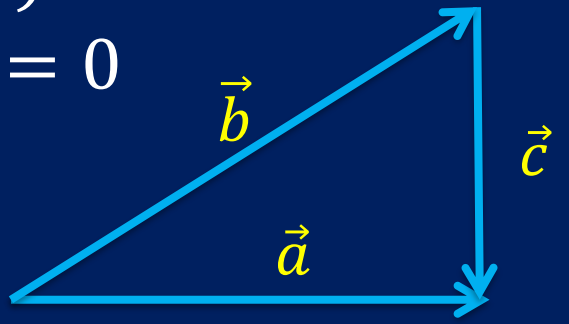
SOLUTION:

We see that $\vec{\mathrm{b}} + \vec{c} = \vec{a}$, therefore given vector form a triangle. Now

$$\vec{a} \cdot \vec{\mathrm{b}} = 3(1) - 2(-3) + (1)(5) = 14$$

$$\vec{\mathrm{b}} \cdot \vec{c} = (1)(2) - 3(1) + 5(-4) = -21$$

$$\vec{a} \cdot \vec{\mathrm{c}} = 3(2) - 2(1) + 1(-4) = 0$$



$\vec{a} \cdot \vec{\mathrm{c}} = 0$ shows that $\vec{a} \perp \vec{c}$

Therefore, the given triangle is right angle triangle.

EXAMPLE:

Show that the vectors $2\hat{\imath} - \hat{\jmath} + \hat{k}$, $\hat{\imath} - 3\hat{\jmath} - 5\hat{k}$, and $3\hat{\imath} - 4\hat{\jmath} - 4\hat{k}$ from the sides of a right-angle triangle.

SOLUTION: Let $\vec{a} = 2\hat{\imath} - \hat{\jmath} + \hat{k}$

$\vec{b} = \hat{\imath} - 3\hat{\jmath} - 5\hat{k}$

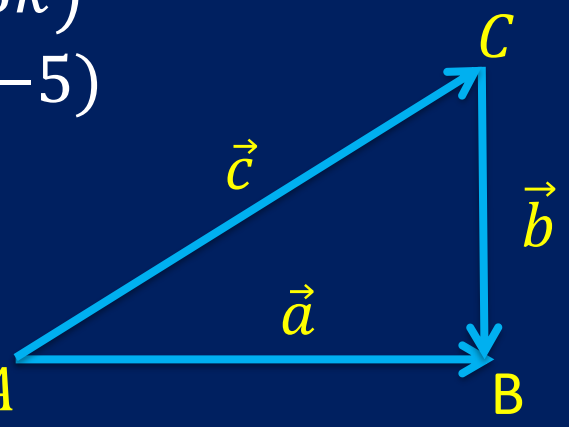
$\vec{c} = 3\hat{\imath} - 4\hat{\jmath} - 4\hat{k}$

Now $\vec{a} + \vec{b} = (2\hat{\imath} - \hat{\jmath} + \hat{k}) + (\hat{\imath} - 3\hat{\jmath} - 5\hat{k}) = 3\hat{\imath} - 4\hat{\jmath} - 4\hat{k} = \vec{c}$

Further prove that Δ is right angle ($\vec{a} \cdot \vec{b} = 0$)

$$\vec{a} \cdot \vec{b} = (2\hat{\imath} - \hat{\jmath} + \hat{k}) \cdot (\hat{\imath} - 3\hat{\jmath} - 5\hat{k})$$
$$= 2(1) + (-1)(-3) + (1)(-5)$$
$$= 2+3-5=0$$
$$\vec{a} \perp \vec{b}$$

Hence Δ ABC is a right triangle.



EXAMPLE: If $\vec{a} \cdot \vec{b} = \vec{a} \cdot \vec{c}$. Show that either $\vec{a} = 0$, or $\vec{b} = \vec{c}$ or $\vec{a} \perp (\vec{b} - \vec{c})$

SOLUTION:

Given that

$$\vec{a} \cdot \vec{b} = \vec{a} \cdot \vec{c}, \text{ therefore}$$

$$\vec{a} \cdot \vec{b} - \vec{a} \cdot \vec{c} = 0$$

$$\vec{a} \cdot \left(\vec{b} - \vec{c}\right) = 0$$

which shows that either $\vec{a} = 0$, or $\vec{b} = \vec{c}$ or $\vec{a} \perp (\vec{b} - \vec{c})$

This implies that either

$$\vec{a} = 0, \text{ or } \vec{b} = \vec{c} \text{ or } \vec{a} \perp (\vec{b} - \vec{c})$$

EXERCISE

Q#1 Given $\vec{a} = -2\hat{i} + 3\hat{j} + \hat{k}$, $\vec{b} = 7\hat{i} + 4\hat{j} + 5\hat{k}$ and $\vec{c} = \hat{i} - 5\hat{j} + 2\hat{k}$. Find the number

a) $\vec{a} \cdot \vec{b}$ (b) $\vec{b} \cdot \vec{c}$ (c) $\vec{a} \cdot (\vec{b} + \vec{c})$

d) $\vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$ (e) $(\vec{a} - \vec{b}) \cdot (\vec{b} + \vec{c})$

Q#2 Find

(a) $\vec{A} \cdot \vec{B}$, $|\vec{A}|$, $|\vec{B}|$

(b) the cosine of the angle b/w $\vec{A}$ and $\vec{B}$

(c) the scalar component of $\vec{B}$ in the direction of $\vec{A}$

(1) $\vec{A} = 2\hat{i} - 4\hat{j} + \sqrt{5}\,\hat{k}$, $\vec{B} = -2\hat{i} + 4\hat{j} - \sqrt{5}\,\hat{k}$

(2) $\vec{A} = \frac{3}{5}\hat{i} + \frac{4}{5}\hat{k}$, $\vec{B} = 5\hat{i} + 12\hat{j}$

(3) $\vec{A} = 10\hat{i} + 11\hat{j} - 2\hat{k}$, $\vec{B} = 3\hat{j} + 4\hat{k}$

(4) $\vec{A} = 2\hat{i} + 10\hat{j} - 11\hat{k}$, $\vec{B} = 2\hat{i} + 2\hat{j} + \hat{k}$

(5) $\vec{A} = 5\hat{j} - 3\hat{k}$, $\vec{B} = \hat{i} + \hat{j} + \hat{k}$

(6) $\vec{A} = -\hat{i} + \hat{j}$, $\vec{B} = \sqrt{2}\,\hat{i} + \sqrt{3}\,\hat{j} + 2\hat{k}$

7) $\vec{A} = 5\hat{i} + \hat{j}$, $\vec{B} = 2\hat{i} + \sqrt{17}\,\hat{j}$

Q#3 Find the angle b/w $\vec{a}$ and $\vec{b}$

1) $\vec{a} = -4\hat{i} + 8\hat{j} - 3\hat{k}$, $\vec{b} = 2\hat{i} + \hat{j} + \hat{k}$

2) $\vec{a} = -2\hat{i} - 3\hat{j}$, $\vec{b} = -6\hat{i} + 4\hat{k}$

3) $\vec{a} = 3\hat{i} - 5\hat{j} - \hat{k}$ $\vec{b} = 2\hat{i} + \hat{j} - 3\hat{k}$

Q#4 Show that $\vec{a}$ and $\vec{b}$ are orthogonal.

(a) $\vec{a} = 3\hat{i} - 2\hat{j} + \hat{k}$, $\vec{b} = 4\hat{i} + 5\hat{j} - 2\hat{k}$

(b) $\vec{a} = 4\hat{i} - \hat{j} - 2\hat{k}$, $\vec{b} = 2\hat{i} - 2\hat{j} + 5\hat{k}$

Q#5 Find all values of c such that $\vec{a}$ and $\vec{b}$ are orthogonal.

(a) $\vec{a} = c\hat{i} - 2\hat{j} + 3\hat{k}$, $\vec{b} = c\hat{i} + c\hat{j} - 5\hat{k}$

(b) $\vec{a} = 4\hat{i} + 2\hat{j} + c\hat{k}$, $\vec{b} = \hat{i} + 22\hat{j} - 3c\hat{k}$

Q#6 If $\hat{u}_1$ and $\hat{u}_2$ are orthogonal unit vectors and $\vec{v} = a\hat{u}_1 + b\hat{u}_2$, find $\vec{v} \cdot \hat{u}_1$

Q#7 Suppose $\vec{A}$, $\vec{B}$ and $\vec{C}$ are mutually orthogonal vectors, Let $\vec{D} = 5\vec{A} - 6\vec{B} + 3\vec{C}$

a) If $\vec{A}$, $\vec{B}$ and $\vec{C}$ are unit vectors find $|\vec{D}|$, the magnitude of $\vec{D}$. Ans $\sqrt{70}$

(b) If $|\vec{A}| = 2$, $|\vec{B}| = 3$ and $|\vec{C}| = 4$ find $|\vec{D}|$

Ans $\sqrt{568}$

Q#8 Suppose $\hat{A}$, $\hat{B}$ and $\hat{C}$ are mutually orthogonal unit vectors. If $\vec{D}$ is a vector such that $\vec{D} = \alpha\hat{A} + \beta\hat{B} + \gamma\hat{C}$, where α, β and γ are scalars, Prove that

$$\alpha = \vec{D}\cdot\hat{A}$$
$$\beta = \vec{D}\cdot\hat{B}$$
$$\gamma = \vec{D}\cdot\hat{C}$$

Q#9 Determine whether the given vectors are orthogonal, Parallel or neither.

(a) $\vec{a} = -5\hat{i} + 3\hat{j} + 7\hat{k}$, $\vec{b} = 6\hat{i} - 8\hat{j} + 2\hat{k}$

(b) $\vec{a} = 4\hat{i} + 6\hat{j}$, $\vec{b} = -3\hat{i} + 2\hat{j}$

(c) $\vec{a} = -\hat{i} + 2\hat{j} + 5\hat{k}$, $\vec{b} = 3\hat{i} + 4\hat{j} - \hat{k}$

Q#10 If $\vec{a}$ is any vector, then Prove that

$$\vec{a} = (\vec{a}\cdot\hat{i})\hat{i} + (\vec{a}\cdot\hat{j})\hat{j} + (\vec{a}\cdot\hat{k})\hat{k}$$

Q#11 Show that vectors $\vec{a} = 3\hat{i} - 2\hat{j} + \hat{k}$, $\vec{b} = \hat{i} - 3\hat{j} + 5\hat{k}$ and $\vec{c} = 2\hat{i} + \hat{j} - 4\hat{k}$ form a right angled triangle.

Also Show that $|\vec{a}|\vec{b} + |\vec{b}|\vec{a}$ and $|\vec{a}|\vec{b} - |\vec{b}|\vec{a}$ are Perpendicular.

Q # 12 Let $\vec{a} = 2\hat{i} - 3\hat{j} + 4\hat{k}$ and $\vec{b} = -3\hat{i} + 4\hat{j} + c\hat{k}$,

Determine c if $\vec{a}$ and $\vec{b}$ are perpendicular.

Ans → 9/2

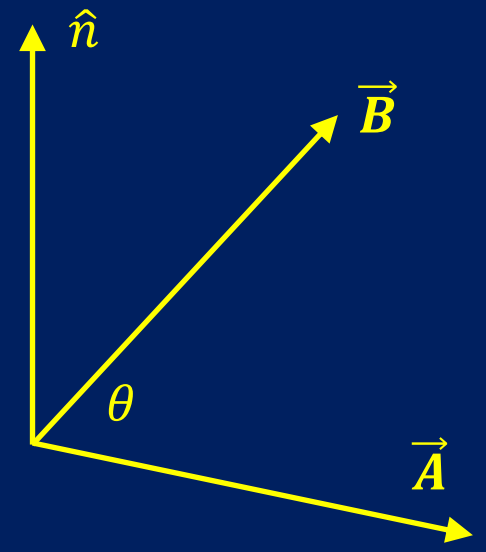
CROSS PRODUCT/ VECTOR PRODUCT

INTRODUCTION

Cross product are widely used to describe the effects of forces in studies of electricity, magnetism, fluid flows and orbital mechanics.

Definition: 1 The vector product or cross product of two vectors $\vec{A}$ and $\vec{B}$ is denoted as $\vec{A} \times \vec{B}$ and is defined as

$$\vec{A} \times \vec{B} = |\vec{A}||\vec{B}| sin\theta \hat{n}$$



Where θ is an angle between $\vec{A}$ and $\vec{B}$

$\hat{n}$ is a unit vector normal to the plane containing $\vec{A}$ and $\vec{B}$ such that the vectors $\vec{A}$, $\vec{B}$ and $\hat{n}$ from a right handed system.

Now by applying the definition of cross product to unit vectors $\hat{\imath}, \hat{\jmath}$ *and* $\hat{k}$ we have

a) $\hat{\imath} \times \hat{\imath} = |\hat{\imath}||\hat{\imath}| \sin 0^0 \, \hat{n} = 0$

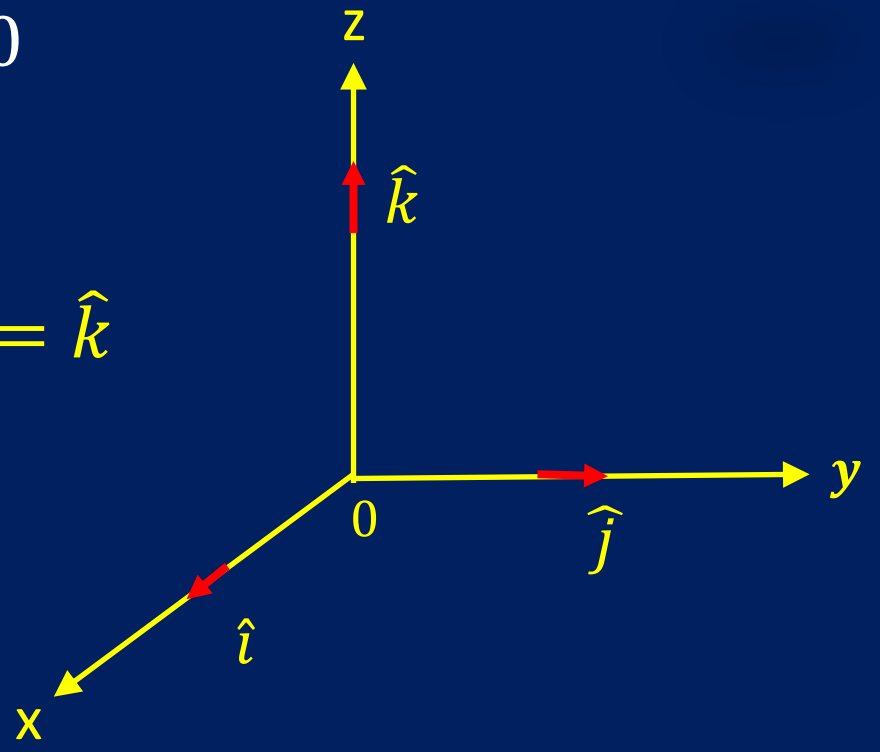
$\hat{\jmath} \times \hat{\jmath} = |\hat{\jmath}||\hat{\jmath}| \sin 0^0 \, \hat{n}$

$\hat{k} \times \hat{k} = 0$

b) $\hat{\imath} \times \hat{\jmath} = |\hat{\imath}||\hat{\jmath}| \sin 90^0 \hat{k} = \hat{k}$

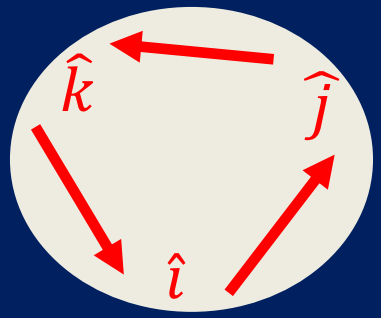
$\hat{\jmath} \times \hat{k} = \hat{\imath}$

$\hat{k} \times \hat{\imath} = \hat{\jmath}$



NOTE:

The cross product of $\hat{\imath}, \hat{\jmath}$ *and* $\hat{k}$ are written in the cyclic pattern.



Definition:2 Determinant formula for $\vec{A} \times \vec{B}$

Our next objective is to calculate $\vec{A} \times \vec{B}$ from the components of $\vec{A}$ and $\vec{B}$ relative to a cartesian coordinate system.

Suppose that

$$\vec{A} = a_1\hat{\imath} + a_2\hat{\jmath} + a_3\hat{k}$$

and $$\vec{B} = b_1\hat{\imath} + b_2\hat{\jmath} + b_3\hat{k}$$

Then

$$\vec{A} \times \vec{B} = (a_1\hat{\imath} + a_2\hat{\jmath} + a_3\hat{k}) \times (b_1\hat{\imath} + b_2\hat{\jmath} + b_3\hat{k})$$

$$= a_1 b_1(\hat{\imath} \times \hat{\imath}) + a_1 b_2(\hat{\imath} \times \hat{\jmath}) + a_1 b_3(\hat{\imath} \times \hat{k})$$

$$+a_2 b_1(\hat{\jmath} \times \hat{\imath}) + a_2 b_2(\hat{\jmath} \times \hat{\jmath}) + a_2 b_3(\hat{\jmath} \times \hat{k})$$

$$+a_3 b_1(\hat{k} \times \hat{\imath}) + a_3 b_2(\hat{k} \times \hat{\jmath}) + a_3 b_3(\hat{k} \times \hat{k})$$

$$= a_1 b_2(\hat{k}) + a_1 b_3(-\hat{\jmath}) + a_2 b_1(-\hat{k}) + a_2 b_3(\hat{\imath})$$

$$+a_3 b_1(\hat{\jmath}) + a_3 b_2(-\hat{\imath})$$

$$= (a_2\, b_3 - a_3\, b_2)\,\hat{\imath} - (a_1\, b_3 - a_3\, b_1)\,\hat{\jmath} + (a_1\, b_2 - a_2\, b_1)\hat{k}$$

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

By interchanging 2nd and 3rd rows

$$\vec{A} \times \vec{B} = -\begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ b_1 & b_2 & b_3 \\ a_1 & a_2 & a_3 \end{vmatrix} = -\vec{B} \times \vec{A}$$

PROPERTIES OF CROSS PRODUCT

1. $\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$ (already proof in determinant formula of $\vec{A} \times \vec{B}$)

2. Scalar Distributive law:

$$(s\vec{A}) \times (r\vec{B}) = rs(\vec{A} \times \vec{B})$$

where **s** and **r** are scalar numbers.

3. Vector Distributive law:

$$\vec{A} \times (\vec{B} + \vec{C}) = \vec{A} \times \vec{B} + \vec{A} \times \vec{C}$$

$$(\vec{B} + \vec{C}) \times \vec{A} = \vec{B} \times \vec{A} + \vec{C} \times \vec{A}$$

4. From definition of cross product

$$\vec{A} \times \vec{B} = |\vec{A}||\vec{B}| sin\theta \hat{n} \quad \text{Eq. (1)}$$

Therefore $$|\vec{A} \times \vec{B}| = |\vec{A}||\vec{B}| sin\theta \quad \text{Eq. (2)}$$

Thus from (1) and (2) $$\hat{n} = \frac{\vec{A} \times \vec{B}}{|\vec{A} \times \vec{B}|}$$ **(which is unit vector)**

Also $$sin\theta = \frac{|\vec{A} \times \vec{B}|}{|\vec{A}||\vec{B}|}$$

5. Parallel Vectors

Non-Zero vectors $\vec{A}$ and $\vec{B}$ are parallel $\Leftrightarrow \vec{A} \times \vec{B} = 0$

Proof If $\vec{A}$ and $\vec{B}$ are Parallel vectors,

so $\theta = 0^0 \Rightarrow sin0^0 = 0.$ then

$$\vec{A} \times \vec{B} = |\vec{A}||\vec{B}|sin\theta\hat{n}$$

$$= |\vec{A}||\vec{B}|sin0^0 = 0.$$

If $\vec{A} \times \vec{B} = 0$ then

$$|\vec{A}||\vec{B}|sin\theta\hat{n} = 0$$

$\Rightarrow$ either $sin\theta = 0$ or $|\vec{A}| = 0$, or $|\vec{B}| = 0$.

1) If $sin\theta = 0 \Rightarrow \theta = 0^0$ or 180^0, which shows that the vectors $\vec{A}$ and $\vec{B}$ are parallel .
2) If $|\vec{A}| = 0$, or $|\vec{B}| = 0$, then since the zero vector has no specific direction. We adopt the convention that the zero vector is parallel to every vector.

6. The vector $\vec{A} \times \vec{B}$ is Orthogonal to both $\vec{A}$ and $\vec{B}$.

Proof: Using (two vectors $\vec{A}$ and $\vec{B}$ are orthogonal $\Leftrightarrow \vec{A} \cdot \vec{B} = 0$)

It is sufficient to show that $(\vec{A} \times \vec{B}) \cdot \vec{A} = 0$ and $(\vec{A} \times \vec{B}) \cdot \vec{B} = 0$

Let $\vec{A} = a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}$, $\vec{B} = b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k}$

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$$= (a_2 b_3 - a_3 b_2)\hat{i} - (a_1 b_3 - a_3 b_1)\hat{j} + (a_1 b_2 - a_2 b_1)\hat{k}$$

$$(\vec{A} \times \vec{B}) \cdot \vec{A} = (a_2 b_3 - a_3 b_2) a_1 + (b_1 a_3 - a_1 b_3) a_2 + (a_1 b_2 - a_2 b_1) a_3$$

$$= a_1 a_2 b_3 - a_1 a_3 b_2 + b_1 a_3 a_2 - a_1 b_3 a_2 + a_1 b_2 a_3 - a_2 b_1 a_3 = 0.$$

Hence $\vec{A} \times \vec{B}$ is Orthogonal to $\vec{A}$

The Proof that $(\vec{A} \times \vec{B}) \cdot \vec{B} = 0$ is similar.

EXAMPLE: Find a vector Perpendicular to each of the vectors $\vec{A} = 2\hat{\imath} - \hat{\jmath} + \hat{k}, \quad \vec{B} = 4\hat{\imath} + 2\hat{\jmath} - \hat{k}$

SOLUTION: A vector perpendicular to both vectors $\vec{A}$ and $\vec{B}$ is $\vec{A} \times \vec{B}$.

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 2 & -1 & 1 \\ 4 & 2 & -1 \end{vmatrix}$$

$$= -\hat{\imath} + 6\hat{\jmath} + 8\hat{k}$$

Verification: $(\vec{A} \times \vec{B}) \cdot \vec{A} = \left(-\hat{\imath} + 6\hat{\jmath} + 8\hat{k}\right) \cdot \left(2\hat{\imath} - \hat{\jmath} + \hat{k}\right)$

$$= -2 - 6 + 8 = 0$$

$(\vec{A} \times \vec{B}) \cdot \vec{B} = \left(-\hat{\imath} + 6\hat{\jmath} + 8\hat{k}\right) \cdot \left(4\hat{\imath} + 2\hat{\jmath} - \hat{k}\right) = -4 + 12 - 8 = 0$

Hence $\vec{A} \times \vec{B}$ is perpendicular to both vectors $\vec{A}$ and $\vec{B}$

GEOMETRICAL REPRESENTATION OF VECTOR PRODUCT

Area of Parallelogram:

The magnitude of the cross product / vector product $\vec{A} \times \vec{B}$ is equal to the area of Parallelogram.

Area of Triangle:

$$Area\ of\ \Delta = \frac{1}{2}\left|\vec{A} \times \vec{B}\right|$$

Proof (Try yourself)

EXAMPLE:

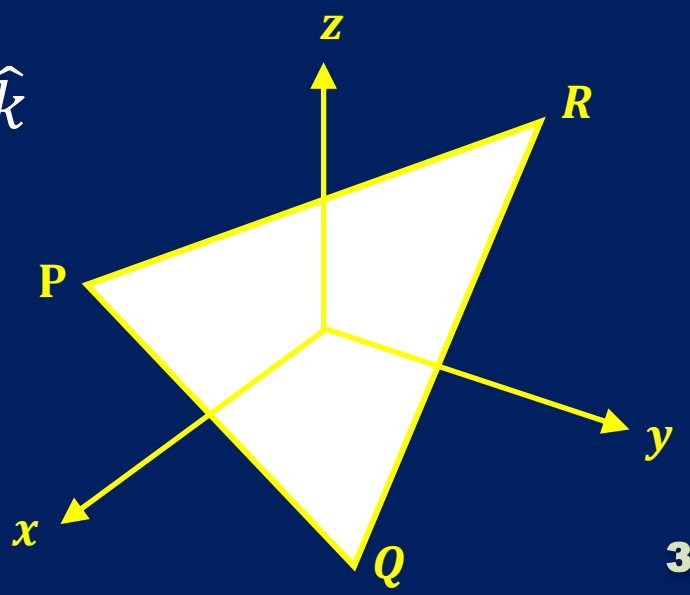
1. Find a vector Perpendicular to the plane of $P(4,-3,1)$, $Q(6,-4,7)$ and $R(1,2,2)$
2. Also find the area of the triangle with vertices $P(4,-3,1)$, $Q(6,-4,7)$ and $R(1,2,2)$.
3. Also find unit vectors perpendicular to plane of P, Q and R.

SOLUTION:

The vector $\overrightarrow{PQ} \times \overrightarrow{PR}$ is perpendicular to plane because it is perpendicular to both vectors

$$\overrightarrow{PQ} = (6\hat{\imath} - 4\hat{\jmath} + 7\hat{k}) - (4\hat{\imath} - 3\hat{\jmath} + \hat{k})$$
$$= (6-4)\hat{\imath} + (-4+3)\hat{\jmath} + (7-1)\hat{k}$$
$$= 2\hat{\imath} - \hat{\jmath} + 6\hat{k} = (2,-1,6)$$

$$\overrightarrow{PR} = (\hat{\imath} + 2\hat{\jmath} + 2\hat{k}) - (4\hat{\imath} - 3\hat{\jmath} + \hat{k})$$
$$= (1-4)\hat{\imath} + (2+3)\hat{\jmath} + (2-1)\hat{k}$$
$$= -3\hat{\imath} + 5\hat{\jmath} + \hat{k} = (-3,5,1)$$



$$\overrightarrow{PQ} \times \overrightarrow{PR} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 2 & -1 & 6 \\ -3 & 5 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} -1 & 6 \\ 5 & 1 \end{vmatrix} \hat{\imath} - \begin{vmatrix} 2 & 6 \\ -3 & 1 \end{vmatrix} \hat{\jmath} + \begin{vmatrix} 2 & -1 \\ -3 & 5 \end{vmatrix} \hat{k} = -31\hat{\imath} - 20\hat{\jmath} + 7\hat{k}$$

Hence area of the parallelogram with adjacent sides P, Q and R is

$$\left|\overrightarrow{PQ} \times \overrightarrow{PR}\right| = \sqrt{(-31)^2 + (-20)^2 + (7)^2} = \sqrt{1410}$$

It follows that area of triangle $= \frac{1}{2}\left(\sqrt{1410}\right) = 18.8$

3) Since $\overrightarrow{PQ}$ and $\overrightarrow{PR}$ is perpendicular to the plane, its direction $\hat{n}$ is a unit vector perpendicular to the plane

$$\hat{n} = \frac{\overrightarrow{PQ} \times \overrightarrow{PR}}{\left|\overrightarrow{PQ} \times \overrightarrow{PR}\right|} = \frac{-31\hat{\imath} - 20\hat{\jmath} + 7\hat{k}}{\sqrt{1410}} = \frac{-31\hat{\imath}}{\sqrt{1410}} - \frac{20\hat{\jmath}}{\sqrt{1410}} + \frac{7\hat{k}}{\sqrt{1410}}$$

EXAMPLE: Find $\vec{A} \times \vec{B}$ and $\vec{B} \times \vec{A}$ if

$$\vec{A} = 2\hat{\imath} + \hat{\jmath} + \hat{k}, \qquad \vec{B} = -4\hat{\imath} + 3\hat{\jmath} + \hat{k}$$

SOLUTION:

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 2 & 1 & 1 \\ -4 & 3 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 1 \\ 3 & 1 \end{vmatrix} \hat{\imath} - \begin{vmatrix} 2 & 1 \\ -4 & 1 \end{vmatrix} \hat{\jmath} + \begin{vmatrix} 2 & 1 \\ -4 & 3 \end{vmatrix} \hat{k}$$

$$= (1 - 3)\hat{\imath} - (2 + 4)\hat{\jmath} + (6 + 4)\hat{k}$$

$$\vec{A} \times \vec{B} = -2\hat{\imath} - 6\hat{\jmath} + 10\hat{k}$$

$$\vec{B} \times \vec{A} = -(\vec{A} \times \vec{B}) = 2\hat{\imath} + 6\hat{\jmath} - 10\hat{k}$$

EXAMPLE: Find a unit vector perpendicular to plane of vector $\vec{a} = 2\hat{\imath} + 2\hat{\jmath} - \hat{k}$ and $\vec{b} = 6\hat{\imath} - 3\hat{\jmath} + 2\hat{k}$

Also find the *sine* of angle b/w $\vec{a}$ and $\vec{b}$.

SOLUTION:

Here

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 2 & 2 & -1 \\ 6 & -3 & 2 \end{vmatrix} = \hat{\imath} - 10\hat{\jmath} - 18\hat{k}$$

Therefore $\left|\vec{a} \times \vec{b}\right| = \sqrt{1 + 100 + 324} = \sqrt{425} = 5\sqrt{17}$

Thus the required unit vector will be

$$\hat{n} = \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|} = \frac{\hat{\imath} - 10\hat{\jmath} - 18\hat{k}}{5\sqrt{17}}$$

Now $|\vec{a}| = \sqrt{4 + 4 + 1} = 3$ and $\left|\vec{b}\right| = \sqrt{36 + 9 + 4} = 7$

$$sin\theta = \frac{|\vec{a} \times \vec{b}|}{|\vec{a}||\vec{b}|} = \frac{5\sqrt{17}}{3(7)} = \frac{5\sqrt{17}}{21}$$